OFFLINE TRANSLATION • QUARANTINED RAG Q&A

Closed-Domain Sanskrit Mahabharata System

Minimal-Footprint Translation Engine & Guardrailed Q&A

1. Executive Summary

This specification details the end-to-end architecture for processing the unabridged Sanskrit Mahabharata using minimal-footprint neural models. The system pairs an offline lightweight translation pipeline with a strictly quarantined Retrieval-Augmented Generation (RAG) Q&A engine. The chatbot is mathematically constrained to answer queries exclusively from the Mahabharata corpus, actively refusing out-of-domain entities such as Ramayana characters, modern pop-culture figures (Jon Snow, Harry Potter), and general world knowledge.

2. Readily Available Datasets

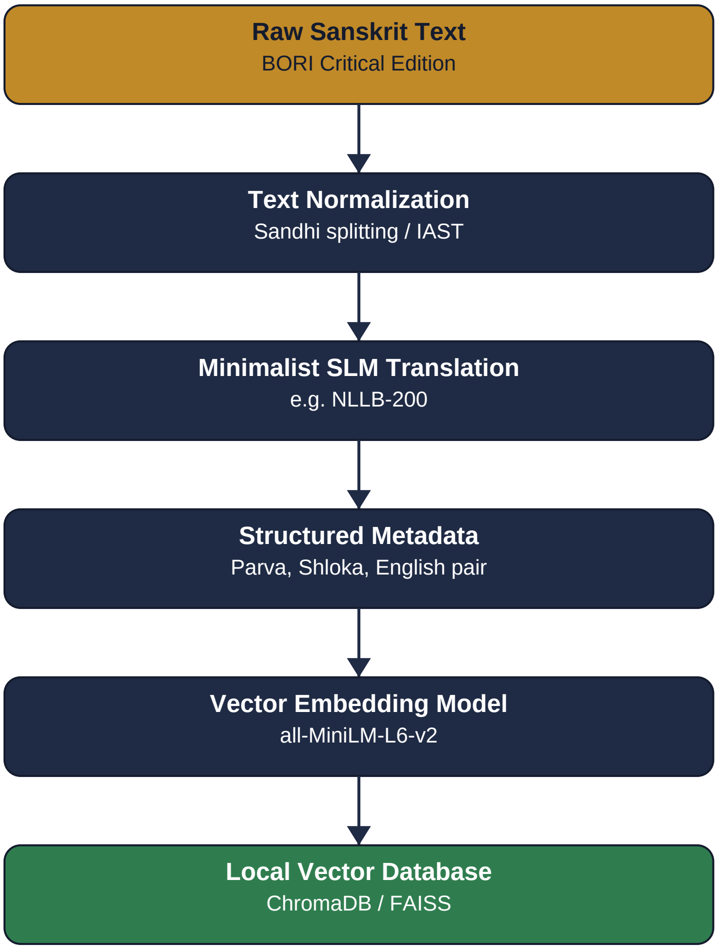
Dataset	Source / Repository	Format & Content	Architectural Role
BORI Critical Edition	GRETIL & SanskritDocuments	18 Parvas, ~80,000 verses in IAST	Primary canonical Sanskrit source text.
K. M. Ganguli Translation	Project Gutenberg	Full English prose translation	Parallel corpus alignment for English semantic grounding.
IIT Bombay Corpus	IITB NLP Repository	Parallel aligned sentence pairs	Base linguistic bootstrapping for the translator model.

3. End-to-End System Pipeline

To eliminate computational overhead and runtime latency, the architecture is decoupled into two independent stages: an **Offline Translation Layer** and an **Online Guardrailed Q&A Layer**.

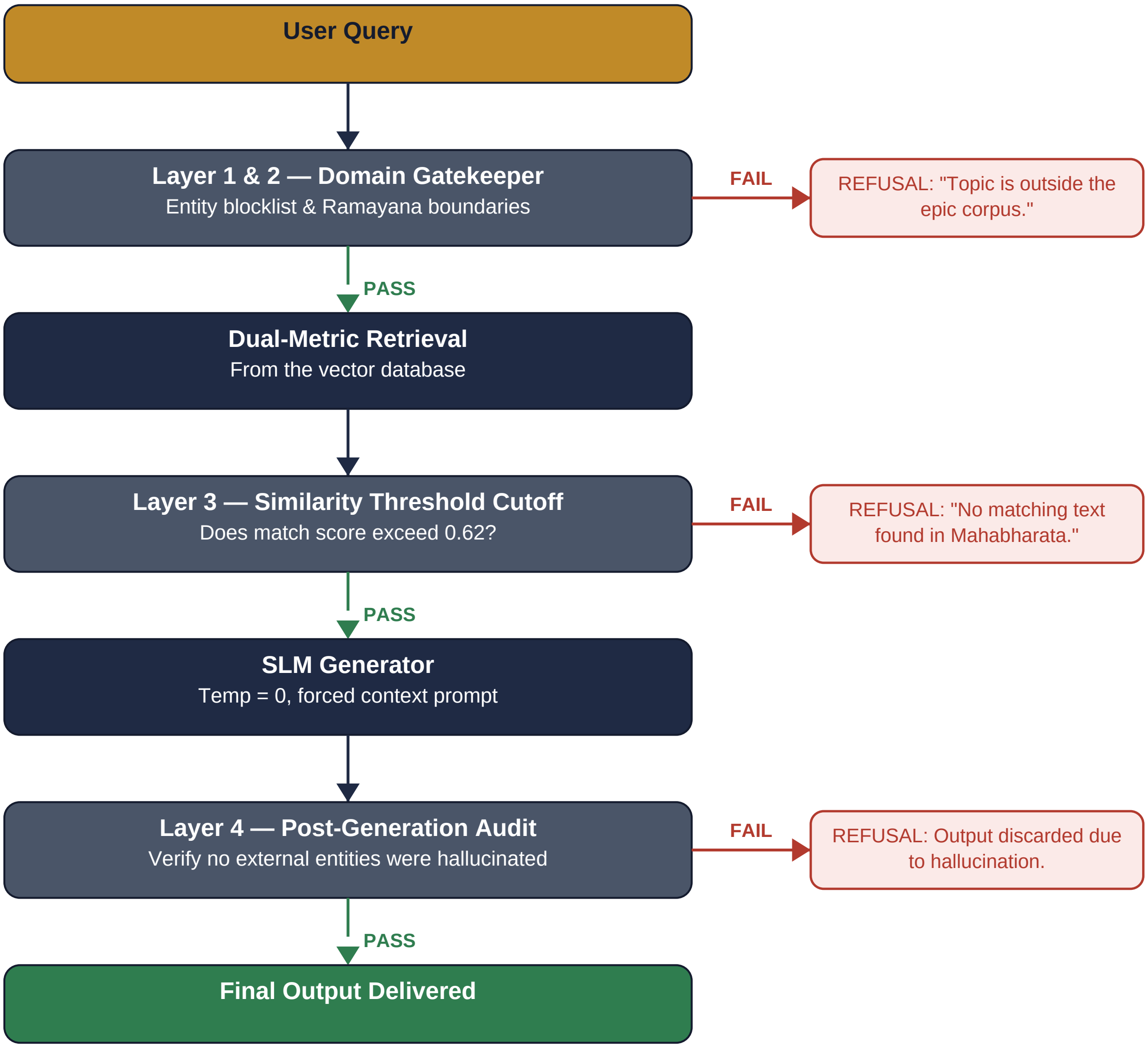
Stage 1 — Offline Ingestion, Translation & Indexing

A one-time execution pipeline that constructs the foundational knowledge base.



Stage 2 — Online Guardrailed Query-Response

A multi-layered defense system that intercepts out-of-domain queries before they ever reach the generator.



4. Ultra-Small Model Selection

System Role	Model Candidate	Size / Footprint	Operational Profile
Sanskrit Translator	NLLB-200 Distilled	600 Million	Runs offline once; zero runtime footprint.
Vector Embedding	all-MiniLM-L6-v2	22M (~90 MB)	Sub-millisecond retrieval speeds.
Domain Gatekeeper	Deterministic Trie Classifier	< 5M Parameters	Pre-retrieval boundary enforcement.
Response Generator	SmolLM2-135M (4-bit)	135M (~120 MB)	Executes strict synthesis on edge devices.

5. Guardrail Behaviour & Test Scenarios

User Query	Active Defense Layer	Decision	Observed Behaviour
“What was Karna's vow?”	Layers 1, 2, 3	Cleared	Synthesizes factual answer with citations.
“Who killed Jon Snow?”	Layer 1 (Entity Gatekeeper)	Refused	Blocked instantly: topic is outside corpus.
“What spells did Harry Potter use?”	Layer 1 & Layer 3	Refused	Fails the vector similarity test.
“Explain Ram vs Ravana.”	Layer 2 (Disambiguation)	Bounded	Restricts response solely to Markandeya's narration in Vana Parva.

*Across every scenario, the same principle holds: **evidence and entity boundaries are checked before generation, and generation is audited after**, so no answer can slip through that isn't grounded in the Mahabharata corpus.*